Where Patients Go When a Hospital Closes: Evidence from Brazil

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June 9, 2026

Abstract

When a hospital closes, the towns that lose access are usually defined by distance: the residents nearest the facility. In public referral networks, where patients cross borders for care, that rule names the wrong towns. Using Brazilian SUS inpatient records, I define each closure's exposed population from revealed pre-closure patient flows. Across 60 closures, the flow and distance rules disagree on 293 of 337 municipality–closure pairs: distance misses the towns that traveled to the hospital and flags nearer ones that never used it. Applying the flow design to 48 psychiatric closures in a staggered event study, the referral link breaks: inpatient psychiatric admissions fall by 69%, a net loss of inpatient psychiatric care rather than a reallocation to other hospitals, which do not measurably absorb the lost inpatient volume; I find no evidence of large acute increases in suicide or self-harm mortality, with the suicide bound excluding increases above roughly one-quarter of baseline—a bounded null that holds across specifications and under a synthetic difference-in-differences design. The design ports to provider-loss settings with administrative data.

Keywords: hospital closures, deinstitutionalization, patient-flow exposure, mortality, suicide, staggered difference-in-differences, Brazil.

JEL: I18, I14, I11, H75.

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1 Introduction

When a hospital closes, which towns lose access? Every closure study has to answer, and the standard answer is distance: residents near the closing facility are treated, those farther away are controls (Avdic, 2016; Carroll, 2019; Joynt et al., 2015; Gujral and Basu, 2019; Lindrooth et al., 2018). The rule is close enough where patients use the nearest facility. It breaks where care runs through referral. In Brazil's Sistema Único de Saúde (SUS), 33.2% of the 179.4 million inpatient admissions between 2010 and 2024 are billed by a hospital outside the patient's home municipality, and the typical municipality sends seven in ten of its admitted residents elsewhere. When patients cross borders this routinely to reach scarce providers, the towns a closure harms are not the ones nearest the building. They are the ones whose residents were going there.

The gap between the two rules is not a rounding error. One psychiatric hospital in the sample drew 2,331 admissions from 20 municipalities before it closed; 16 of them sat between 36 and 107 kilometers away, a median of 44 km. Any distance cutoff tight enough to be credible would have called those towns unexposed, though they were precisely the ones that lost their provider. A distance rule and a flow rule do not estimate noisier versions of the same effect; they estimate provider loss for different populations, and if the exposed set is wrong every downstream coefficient inherits the error. Figure 1 shows the mismatch is national: the municipalities whose residents travel farthest beyond their nearest hospital cluster in the river-corridor North and the semi-arid Northeast.

The paper's contribution is a portable way to measure who a closure exposes; the mortality study is where I take that measure to show the stakes are real. For each closure I rebuild origin–destination patient flows from the pre-closure years and define the exposed catchment as the municipalities that sent at least 5% of their SUS admissions to the closing hospital—a rule any single-payer or claims-rich system can build from records it already keeps. The measurement question, whether reading exposure off flows rather than distance changes who is treated, uses all 60 economically meaningful closures (2012–2018 and 2022–2023). The application takes the rule to where flow and distance pull apart most: the 48 psychiatric closures anchored to federal PNASH inspection cycles. Care there concentrates in distant referral hubs, and the welfare concern—suicide and self-harm after the loss of inpatient psychiatric care—matches the shock. I ask whether the provider loss raised mortality in a staggered event study (Sun and Abraham, 2021), with closure timing plausibly orthogonal to local mortality shocks and pre-periods observed back to 2010.

The two rules disagree sharply. Across the 60 closures, flow and distance part ways on 293 of the 337 municipality–closure pairs that either flags; only 44 are caught by both. Distance misses 97 pairs that depended on the closing hospital through long-distance referral and falsely flags 196 that never used it. The errors are not random across space; they concentrate where referral travel is longest (Figure 2). A first stage confirms the flow rule catches real ties rather than noise: in the psychiatric sample, inpatient travel falls by 7.1 km after closure among flow-exposed municipalities, with no pre-trend ($p = 0.96$), as the long trips to the closing hospital disappear.

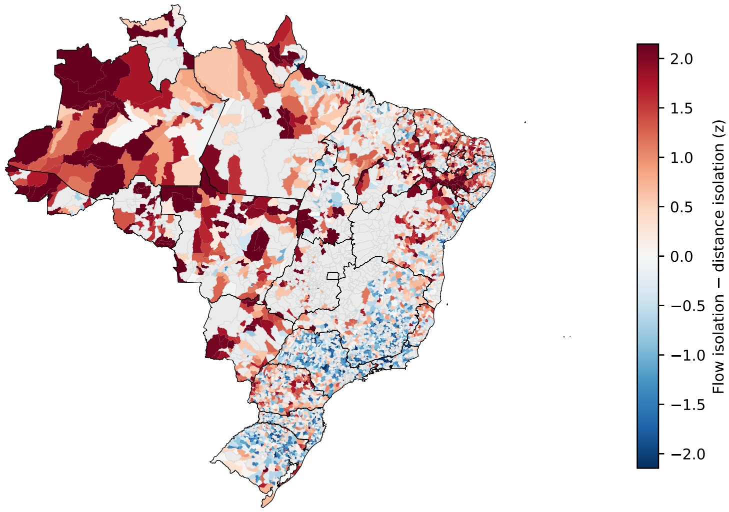

Figure 1: **The hospital people use is often farther than the hospital next door.** Each municipality is shaded by its flow isolation—the admission-weighted distance to the hospitals its residents actually used—minus its distance isolation, the kilometers to the nearest SUS hospital. Red municipalities are more isolated by realized flow than by geography (residents bypass the nearest facility for distant referral) and cluster in the river-corridor North and semi-arid Northeast; blue municipalities are the reverse. Grey municipalities have no recorded SUS admissions. $N = 4{,}429$ municipalities.

The closures bite. Inpatient psychiatric admissions among exposed municipalities fall by 69% (-7.32 per 1,000, 95% CI $[-11.38, -3.27]$). To confirm this is a net loss rather than a reallocation, I trace where the closing hospital's inpatient volume goes: admissions to other hospitals in the catchment do not rise to replace it—a staggered event study puts the change at 0.08 per 1,000 (95% CI $[-0.74, 0.91]$), an unchanged null, against the 69% fall in total admissions—so the headline decline is a net loss of inpatient psychiatric care, not a redistribution across providers.

Against a shock that large, the mortality estimates remain close to zero and imprecise. The population-weighted suicide estimate is +0.28 per 100,000 (95% CI $[-0.72, +1.29]$), against a treated-group baseline of 5.01, with flat pre-trends ($p = 0.26$); self-harm mortality is likewise imprecise around zero. The interval rules out increases above roughly a quarter of baseline suicide, on the order of 38 additional deaths a year across the exposed catchments, but it cannot resolve smaller harms or non-fatal welfare losses. This is a bounded null, not a precise zero, and at 48 closures I do not pretend otherwise. It is not, however, an artifact of one specification: the null holds across 15 analytic choices, in a stricter exact ±1-year PNASH-window sample, and under a synthetic difference-in-differences design that does not rest on never-treated parallel trends (Figure 5). Read narrowly, the closure side of Brazil's psychiatric reform did not produce the acute mortality spike its critics feared, on the suicide margin and at the precision these data allow.

One outcome runs the other way, and I decline to read it causally. Preventable hospitalizations

(ICSAP), the canonical Brazilian access-to-care marker (Macinko and Harris, 2015; Rocha and Soares, 2010), rise by +1.99 per 1,000 after closure. At face value that looks like access deterioration. It is not identified: ICSAP was already climbing steeply in the exposed municipalities for years before any hospital closed, the parallel-trends test rejects decisively ($p < 0.001$), and a Rambachan–Roth sensitivity confirms that no bounded extrapolation of the pre-trend rescues an effect. I report the trajectory in full and treat it as a description of where these towns were already heading, not as a result.

The paper speaks first to the JHE literature that reads hospital closures as access shocks. Buchmueller et al. (2006) ask how closures move access and mortality through distance to the nearest hospital, and Avdic (2016) traces the mortality consequences of the longer trips that emergency-hospital consolidation forces; related work follows closures across the United States (Carroll, 2019; Joynt et al., 2015; Gujral and Basu, 2019; Lindrooth et al., 2018) and hospital entry and exit as swings in local capacity (Petek, 2022). These designs read exposure off geographic proximity, which holds where local capacity is thick. I move the access margin upstream—before asking whether distance to care affects outcomes, I ask whether distance identifies the exposed population at all when patients cross borders for referral. The object can even flip sign from the standard case: shutting a specialized referral provider may *reduce* measured travel, because the long inpatient trip to it disappears, so what marks exposure is revealed reliance on the closing provider, not proximity to the nearest one.

A second, health-services literature uses patient flows to delineate hospital service areas and referral regions, from Wennberg and Gittelsohn (1973), whose small-area flow analysis seeded the Hospital Referral Regions the Dartmouth Atlas has mapped since, to recent evidence that even U.S. Medicare flows drift from those lines (Wang et al., 2020); Brazil's 33.2% cross-border share is larger still. I put the revealed-utilization logic to a different use. Rather than ask what the natural hospital market is, I use pre-closure flows to assign treatment in a provider-loss design—the exposed population is the set of municipalities whose residents actually used the provider before it closed—and post-closure flows to ask whether that inpatient use reappears elsewhere. My point is not that distance is always wrong, but to show, across 60 real closures, how wrong it can be, where, and in which direction, with a flow-based correction built from records these systems already keep.

The paper speaks, finally, to the economics of deinstitutionalization, where its closest companion is Dias and Fontes (2024). They study the reform's other half, the rollout of community CAPS centers, and find that the expansion enlarged outpatient care and cut psychiatric hospitalizations without lowering mental-health mortality. That is the demand side. This paper takes the supply side: the closures that removed inpatient capacity, among the towns revealed to depend on it. In a rollout design treatment falls on the municipality that receives the new facility; in a closure design the affected population is not the host municipality, nor its neighbors, but the set of places that relied on the provider before it shut. Read together, the two papers separate two margins of the reform—the expansion of community care and the loss of inpatient providers; both pull psychiatric inpatient use down, and neither shows large mental-health mortality effects at the margins studied. The design also ports beyond psychiatry: any provider-loss setting with administrative utilization data (rural obstetric units, emergency departments, specialty clinics) can be analyzed the same way, and the general-hospital closures show the exposure problem is not unique to psychiatric reform.

2 Institutional setting and closure samples

The question this paper answers is whether losing a particular inpatient provider changed acute mortality in the places that depended on it. Both the question and the measurement problem inside it come from three features of Brazilian health care. Patients travel across municipal lines for hospital care on a scale that makes geographic proximity a weak guide to who relies on a given hospital. The psychiatric reform that dominates the closure record shut specialized hospitals on a federally driven schedule, and that schedule is what makes closure timing usable for identification. The closures themselves, read off the federal establishment register, define the sample. I take the three in turn.

2.1 SUS regionalization and cross-border patient flow

Brazil's Sistema Único de Saúde (SUS), created by the 1988 Constitution and built out by Laws 8.080 and 8.142 of 1990, is a tax-financed system meant to cover the entire resident population. About 71% of Brazilians use it as their only source of care, and another 25% use it alongside private insurance (Macinko and Harris, 2015). Inpatient care runs through a mix of municipal, state, philanthropic, and contracted private hospitals, arranged in a nested geography. Each of Brazil's 5,570 municipalities is a management unit; municipalities group into 438 Regiões de Saúde (CIRs) meant to coordinate flows across levels of care; and the 27 Federation Units operate or contract the referral hospitals at the top.

Flow does not follow geography. The architecture prescribes a regional hierarchy. It does not enforce one. Patients cross municipal and CIR lines as a matter of routine, most of all for specialized care. Of the 179.4 million SUS admissions in the pooled patient-flow graph (2010–2024), 33.2% are recorded at a hospital outside the patient's home municipality on a volume-weighted basis; weighting municipalities equally, the mean outflow share is 70%, which says that small municipalities lean on care elsewhere far more than the national volume figure lets on. In Sergipe, Pernambuco, and the Distrito Federal the cross-border rate tops 55%. These trips are not noise around a local-care norm. They are how much of the SUS population reaches a hospital bed at all, and they are what breaks the distance proxy: a patient whose usual referral hospital sits one CIR over is coded unexposed to its closure under a nearest-hospital rule, even though that hospital is where she actually went. The distance proxy standard in the closure literature (Avdic, 2016; Carroll, 2019; Joynt et al., 2015; Lindrooth et al., 2018) assumes flow tracks road geography. That assumption holds up where local capacity is thick, as in Sweden or much of the NHS, and is shakier even in U.S. Medicare, where Wang et al. (2020) find patient flows drifting away from the Hospital Referral Region lines the Dartmouth Atlas has drawn since the early 1990s. Brazil's 33.2% runs larger than any of these, and it is exactly where a flow-based correction should bite hardest.

2.2 The Reforma Psiquiátrica and the source of closure timing

The closure record is dominated by a long-running overhaul of mental-health care, and that overhaul is both where the identifying variation comes from and why the mortality question is politically loaded. Lei 10.216 of 2001 mandates deinstitutionalization: asylum-style psychiatric hospitals (*hospitais psiquiátricos*) are to give way to community outpatient networks (Centros de Atenção Psicossocial, CAPS) and supported residences (Serviços Residenciais Terapêuticos). Dias and Fontes (2024) study the CAPS rollout and find it expanded outpatient care and cut psychiatric hospitalizations without lowering mental-health mortality. This paper takes the other side of the same reform, the closures, and asks what happened in the catchments that had relied on those inpatient beds. Two federal levers drove the closures, and they reinforced each other.

PNASH inspection cycles. The Programa Nacional de Avaliação dos Serviços Hospitalares (PNASH) inspected every accredited psychiatric hospital in waves—2002–2003, 2007–2008, 2011–2012, and 2015–2016—scoring infrastructure, staffing, and care processes. A facility scoring below roughly 61% on the composite index drew an indication for de-accreditation under Portaria SAS/MS 501 of 2007. The schedule is the reason to expect closure timing to be loosely tied, at most, to municipality-specific mortality shocks: the inspection calendar is set in Brasília, not chosen locally in response to a town's demand or supply conditions. The primary sample anchors to that calendar (closures within ± 1 year of a cycle; Section 4.2).

Reimbursement erosion. The SUS daily rate for psychiatric inpatient care sat near R$30 per patient-day from the early 2000s and was never indexed to inflation; across the panel window its real value fell by about half. The squeeze pushed facilities into insolvency at scale, again on a clock set by federal reimbursement policy rather than by any single municipality's finances or health needs.

Both levers apply the same federal thresholds and the same reimbursement schedule to all 5,570 municipalities, and both produce decisions at the level of the institution, not the municipality. None of this makes closure timing random. It does make a municipality's own health shocks a less natural explanation for when a psychiatric hospital closed than the PNASH cycle and the reimbursement squeeze. The design therefore treats timing as plausibly orthogonal to local shocks, conditional on the PNASH-anchored sample, the pre-trend checks, and the robustness exercises that follow. Cross-referencing the closures against the homologated PNASH results bears out the mixed-driver picture: of the 48 psychiatric closures, 6 were formally indicated for de-accreditation in the 2012–2014 cycle (Portaria SAS/MS 1.727/2016), while 9 had been classified as adequate and closed anyway—closure here is driven by reimbursement pressure alongside, not only by, quality de-accreditation. I also classified each event's declared closure motive from documentary sources manually. That classification organizes robustness sub-samples and is reported in the replication notes, but its external validation is incomplete, so it does no identifying work.

2.3 Closure events, 2010–2024

Closures come from the Cadastro Nacional de Estabelecimentos de Saúde (CNES), the federal establishment register. I flag every establishment whose reporting stops before the window ends. The raw list is mostly registry debris—small clinics, day units, code reassignments—so a cumulative filter (F1–F5; online appendix) pares it to 60 closures that clear five conditions: closure inside the non-pandemic windows 2012–2018 or 2022–2023; a hospital-grade establishment type (`tp_unid` $\in \{05, 07, 15, 62\}$); at least 30 SUS-contracted beds and at least 100 admissions in the year before closure; and no admission drop above 50% in that final year, which screens out hospitals that had already emptied out under falling demand. Of the 60, 41 are specialized hospitals (`tp_unid` $= 07$, mostly psychiatric), 18 are general (`tp_unid` $= 05$), and one is a day unit. The Southeast accounts for 62% of the events and the Northeast for 25%; the busiest year is 2012, with 11 events, an echo of the early audit cycles.

The causal sample is narrower than the measurement sample. The mortality analysis does not use all 60 closures. It uses the PNASH-anchored psychiatric subset, where the timing argument is strongest and where the suicide and self-harm margins line up with the nature of the shock. The primary causal sample is the 48 psychiatric closures falling within ± 1 year of a PNASH cycle; the 41 statistically filtered specialized closures and a wider 60-closure psychiatric set serve as robustness checks (Section 6). The hand-coded motive labels enter only as additional robustness sub-samples and are cross-tabulated against facility type in the replication notes. They document the events; they do not certify any one of them as causal.

2.4 Four samples, four jobs

The paper leans on four samples, each for a different claim, and keeping them apart matters for reading the results. The F5 sample of 60 economically meaningful closures is the measurement sample: it carries the flow-versus-distance comparison and the general first-stage diagnostics. The 48 PNASH-anchored psychiatric closures are the causal sample for suicide and self-harm. The 18 general-hospital closures and the lone hospital-day closure are contrast cases, used only to ask whether flow-defined catchments matter outside psychiatric reform. Closures that pass the bed and volume screens but fail F5 on a large pre-closure decline are diagnostic cases, held out of the main design because for them the official closure date is a poor stand-in for an abrupt loss of inpatient care. Table 1 sets out the four and the role each one plays.

Table 1: Closure samples and their role in the paper

Sample	Definition	Closures	Flow-exposed munis	Role
F5 measurement sample	Economically meaningful hospital closures passing F1–F5	60	130	Measurement: flow versus distance
PNASH psychiatric causal sample	Psychiatric closures within ±1 year of PNASH cycles	48	104	Causal mortality analysis
General-hospital contrast	F5 closures with general-hospital type	18	24	Contrast evidence
Hospital-day contrast	F5 closures with hospital-day type	1	1	Contrast evidence
Failed-F5 diagnostic sample	Pass F4 but fail F5 due to pre-closure volume decline	30	26	Endogeneity/filter diagnostic

Notes: F5 is the measurement-sample filter for economically meaningful hospital closures. The PNASH psychiatric causal sample prioritizes institutional anchoring to federal inspection cycles; stricter and broader psychiatric samples are reported as robustness checks.

3 Data

The unit of analysis is the municipality-year, and the panel covers the universe of Brazilian municipalities, their public hospital network, inpatient admissions (2010–2024), and mortality (2010–2023). Four federal sources supply the data. The hospital admissions register (SIH-RD, 3.1) yields both the patient-flow exposure measure and the travel-burden outcome. The vital registration database (SIM, 3.2) yields the mortality outcomes. The establishment register (CNES, 3.3) yields the closure events. IBGE auxiliary panels (3.4) supply population denominators and municipality centroids. The documentary sources behind the robustness sub-samples are described in the replication notes. Table 2 reports summary statistics by macro-region.

Two regional patterns shape the design from the outset. Travel burden is structurally unequal: a resident of the Norte or Centro-Oeste averages 94–96 km from home to the admitting hospital, more than double the 42–45 km of the Sudeste and Sul. And the outflow share—the fraction of a municipality's SUS admissions sent outside its own borders—is high everywhere, between 58 and 73%. The second pattern is what lets a closure-driven design operate on real referral dependence rather than noise. The first is why it must: the same cross-region heterogeneity confounds any cross-sectional comparison of access and health, so the paper identifies effects only from within-municipality variation under a closure shock.

3.1 SIH-RD: admissions, the flow graph, and travel burden

The Sistema de Informações Hospitalares–Reduzido (SIH-RD) is the billing record for every SUS-financed inpatient admission, one record per Autorização de Internação Hospitalar (AIH). The window runs 2010–2024 at monthly granularity and holds 179.4 million admissions, linking municipalities of patient residence (`MUNIC_RES`) to roughly 7,000 active SUS hospitals (`CNES`). It sees the SUS-using population only—about 96% of residents—and not the private inpatient flows of the rest, so the estimates are identified within that population (Section 7.4).

From this microdata I build the weighted bipartite graph whose edges (m, h, t) count admissions of residents of m at hospital h in year t. That graph defines the flow exposure shares of equation (1), and two outcomes derive from it. *Travel burden* is the admission-weighted haversine distance from m's centroid to the centroid of the municipality of admission, measured at the municipality-year level; it is the first-stage outcome that tests whether exposure is picking up realized utilization dependence rather than mere proximity (Section 5.2). *ICSAP* counts admissions whose primary diagnosis (`DIAG_PRINC`) falls in the Brazilian primary-care-sensitive list (Portaria MS 221/2008) per 1,000 population. It is the canonical Brazilian access marker (Macinko and Harris, 2015; Rocha and Soares, 2010), and I carry it openly as a non-identified outcome (Section 5.6).

3.2 SIM: psychiatric mortality outcomes

The Sistema de Informação sobre Mortalidade (SIM) records every death in Brazil with cause coded in ICD-10; the window holds 18.9 million deaths over 2010–2023, each attributed to the municipality of *residence* (`CODMUNRES`). The reform's critics warned about one thing in particular: deaths among discharged or displaced psychiatric patients. I build two residence-based outcomes aimed straight at that fear. *Suicide* mortality counts deaths from intentional self-harm (ICD-10 X60–X84) per 100,000. A broader *self-harm* measure adds deaths of undetermined intent (Y10–Y34), which soak up the routine misclassification of suicides as accidents in vital registration. I also follow deaths with a mental or behavioural disorder as the underlying cause (F00–F99) as an auxiliary series, but I do not read it as a welfare outcome: such deaths partly reflect in-facility mortality, which falls mechanically when an asylum shuts its doors.

Why residence-based recording is the right choice here. Attributing deaths to residence rather than to place of occurrence is not a detail. A displaced patient who dies after traveling for care—or after failing to travel—is counted against her home municipality, which is the unit the exposure rule treats. The occurrence-based attribution that SIM also carries would instead assign such a death to the hospital's municipality, severing the link to the exposed population, so I use the residence field throughout. Two limits of SIM survive that choice. Ill-defined-cause recording (R96–R99) is heavier in the rural North and Northeast (14–22%, against 4–6% in the South), which adds noise to mortality rates there; and suicide is under-recorded relative to undetermined-intent deaths, which is the reason the broader self-harm measure travels alongside it. Municipality fixed effects in the event study absorb both to the extent they hold steady over time.

3.3 CNES: establishments, beds, and closure detection

The Cadastro Nacional de Estabelecimentos de Saúde (CNES) is the monthly federal register of every health establishment. Its establishment file (`cnes_st`) gives establishment type (`tp_unid`) and municipality; its bed file (`cnes_lt`) gives SUS-contracted beds (`QT_SUS`), which feed the closure filter. I detect a closure when a `CNES` stops reporting to `cnes_st` after a candidate year. The candidate set—3,518 establishments that cease reporting between 2010 and 2023—is dominated by registry artefacts, so the cumulative F1–F5 cascade (online appendix) trims it to the 60 F5 measurement-sample closures: 41 specialized hospitals, 18 general hospitals, and one hospital-day unit (Section 2.3). The cascade does not catch every false positive. The hand-coded closure-motive classification in the replication notes documents the events that survive and defines the robustness sub-samples.

3.4 IBGE auxiliary: population and centroids

Population denominators come from the IBGE annual estimates (SIDRA table 6579) and the 2022 Census. One construction choice carries weight for the pre-trend test. The rate outcomes need a denominator in every panel year, but the early cohorts' pre-treatment years fall before 2015. I therefore extend the denominators back to 2010, using the IBGE annual estimates for 2011–2014 and a linear back-extrapolation for 2010, so that every cohort's pre-treatment trajectory is observed rather than missing. The stakes are concrete: without the backfill the early cohorts' pre-period is empty and the parallel-trends test says nothing; with it, the test has content, and it is precisely what exposes the pre-existing ICSAP trend (Section 5.6). Municipality centroids (IBGE shapefile, 2022 vintage) enter the haversine distance behind the distance exposure rule E2 and the travel-burden outcome.

The centroid-to-centroid distance approximation. CNES does not record hospital coordinates, so I compute distances centroid to centroid between the patient's municipality and the admitting hospital's municipality, as the Brazilian access literature does (Macinko and Harris, 2015; Rocha and Soares, 2010). The approximation understates within-municipality distance in large urban municipalities and in near-border cases, which attenuates the distance measure. Dropping the 27 state capitals, where hospital dispersion is greatest, moves the first-stage travel ATT by less than 1% (Section 6), and a coordinate-level sensitivity for a sub-sample of closures appears in the online appendix.

Table 2: Summary statistics by Brazilian macro-region, 2015–2023 pooled.

Region	Munis	Pop (med, k)	GDP (med, R$M)	Travel (km)	ICSAP per 1k	Outflow share
Norte	449	16.8	0.2	95.7	12.6	0.58
Nordeste	1,794	14.4	0.1	69.1	12.4	0.73
Centro-Oeste	466	9.5	0.3	94.3	13.5	0.62
Sudeste	1,668	11.3	0.3	42.3	13.6	0.71
Sul	1,188	7.2	0.3	45.0	18.9	0.73
National	5,565	11.5	0.2	59.9	14.3	0.70

Notes: Cells are municipality-level statistics averaged across 2015–2023. *Pop* is the median municipality population in thousands; *GDP* is the median municipal GDP in millions of R$ (constant); *Travel* is the average weighted-mean centroid-to-centroid haversine distance from residence to admission hospital; *ICSAP per 1k* is the average rate of primary-care-sensitive hospitalizations per 1,000 inhabitants (Portaria MS 221/2008 list); *Outflow share* is the municipality-mean share of admissions sent outside the residential municipality (the volume-weighted national counterpart, 33.2%, is reported in Section 2.1). Source: SIH-RD 2010–2024; IBGE 2015–2024 population and 2010–2023 municipal GDP panels.

4 Measuring exposure to hospital closures

Everything in the paper turns on two choices: who counts as exposed to a hospital closure, and what variation identifies the effect. The first is a measurement choice, the patient-flow exposure rule that replaces distance (Section 4.1). The second is the timing of closure, together with the institutional argument for treating it as plausibly orthogonal to local shocks and the three assumptions that argument rests on (Section 4.2). Section 4.3 gives the estimator. The object never changes: the effect of a closure on the populations that depended on the closing hospital, and the contribution is getting that population right.

4.1 Patient-flow exposure

The estimand is not the effect of gaining community mental-health capacity, nor the average effect of Brazil's psychiatric reform. It is the effect of losing a specific inpatient provider for municipalities that demonstrably relied on it before it closed. The unit of exposure is a closure-specific catchment, not the hospital's host municipality and not a radius drawn around the building.

The flow rule (E1). For each closing hospital h with closure year $t_h^\star$, let

$$s_{m,h} \;=\; \frac{w_{t_h^\star-1}(m,h)}{\sum_{h'\in H_{t_h^\star-1}} w_{t_h^\star-1}(m,h')} \tag{1}$$

be the share of municipality m's SUS admissions in the year before closure that went to h, where $w_t(m,h)$ counts admissions from residents of m to hospital h in year t, read off the bipartite municipality–hospital graph of 179.4 million admissions (Section 3). Municipality m is *flow-exposed* to the closure of h when $s_{m,h} \geq \theta$, with default $\theta = 0.05$. The rule reads dependence off care-seeking itself: a municipality whose residents systematically traveled to h is exposed even when h is far away, and a hospital next door that residents never used is left out, as it should be. A continuous sweep $\theta \in [0.01, 0.20]$ in Section 6 shows the result does not hinge on the cutoff.

Distance benchmarks (E2). The comparison is the rule standard in the closure literature: m is *distance-exposed* to h if h is among the three kilometer-nearest hospitals to m's centroid in the pre-closure year,

$$m \in \mathrm{E2}(h) \iff h \in \operatorname{argmin}^{(3)}_{h'\in H_{t_h^\star-1}} \mathrm{km}(m,h'), \tag{2}$$

where top-3 is a deliberately generous catchment, so the comparison set is not made small by construction. Generic distance is a weak yardstick for psychiatric closures, so I add specialty-aware benchmarks too: the three nearest psychiatric providers, the three nearest hospitals of the same CNES establishment type, and the single nearest psychiatric provider. These are classification benchmarks, not estimands I prefer. They ask a sharper question than generic distance does: can geography recover revealed reliance once specialty is taken seriously?

Where the two rules disagree. The two rules are not nested. In the F5 measurement sample of 60 economically meaningful closures, of the 337 municipality–closure pairs that either rule flags, only 44 are flagged by both. 97 are flow-only—residents depended on h, but it was not among their nearest hospitals—and 196 are distance-only—h sits in the nearest-hospital catchment, but residents never used it. Distance misclassifies 293 of the 337 pairs. Table 3 extends the comparison to the specialty-aware benchmarks for the PNASH psychiatric causal sample under same-state risk sets. Generic distance does poorly, and specialty-aware geography helps only where the support is well defined; some psychiatric-provider rules end up flagging nearly the entire support, so I report them as support failures rather than as classifiers. The first-stage travel-burden result (Section 5.2) then shows that the flow-exposed municipalities are precisely the ones whose realized inpatient travel moves when h closes.

Table 3: Patient-flow exposure versus distance-based exposure

Sample / benchmark	Risk set	Flow only	Distance only	Both	Union	Jaccard
F5 measurement sample: top 3 any hospital	National union	97	196	44	337	0.131
PNASH: Top 3 any hospital	Same state	82	91	24	197	0.122
PNASH: Top 3 same type	Same state	38	863	68	969	0.070

Notes: The first row reports the paper's full F5 measurement-sample result: distance misclassifies 293 of 337 municipality–closure pairs flagged by either rule. The PNASH panel restricts distance comparisons to same-state risk sets to avoid national support artifacts. Two narrow specialty-provider rules (nearest and top-three psychiatric provider) classify nearly the entire risk set as exposed—a support failure of naïve specialty geography rather than a usable classifier—and are reported with the full risk-set results in the online appendix.

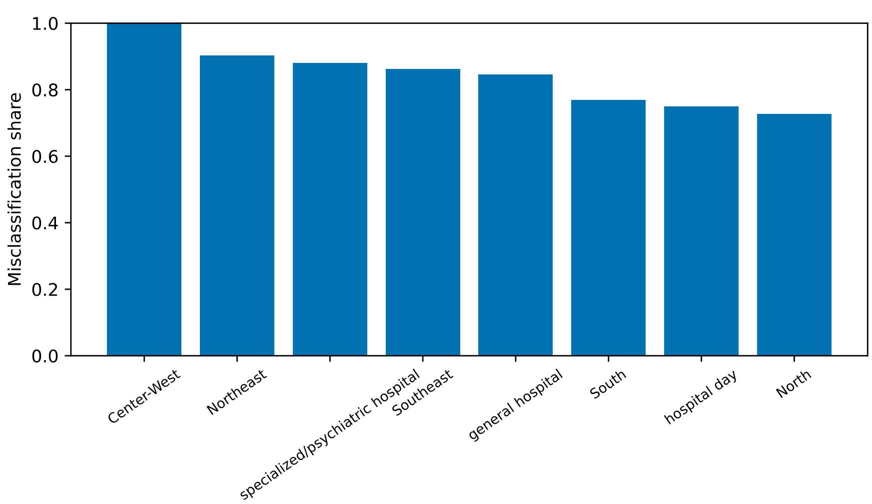

Figure 2: Flow-distance disagreement in the full F5 measurement sample. Bars report the share of municipality–closure pairs flagged by exactly one exposure rule, by hospital type and macroregion.

The embedding is a diagnostic, not the treatment. The replication notes report node2vec diagnostics of the referral network, and they are descriptive only. Treatment assignment is the share-based E1 rule of equation (1), which does not depend on the embedding or on its random seed.

4.2 Why mortality estimation focuses on PNASH psychiatric closures

What the design asks of the data. The flow-versus-distance comparison uses the full F5 sample; the causal mortality estimates use the PNASH psychiatric subset. The design does not require psychiatric closures to be randomly assigned, and I do not claim they are. The flow share $s_{m,h}$ is the equilibrium of a referral network built up by past policy and patient sorting; it is not the identifying variation. The identifying variation is the closure event $h \to \emptyset$ at year $t_h^\star$. What identification requires is that closure *timing*, conditional on the PNASH-centered sample and the municipality and year fixed effects, not coincide with unobserved shocks to mortality in the exposed catchments. The institutional case for that condition is concrete. Both federal levers that drove the closures ran on calendars set in Brasília: the PNASH inspection waves, which scored every accredited psychiatric hospital against a national de-accreditation threshold, and the unindexed SUS daily rate, which lost about half its real value over the panel and pushed facilities into insolvency. Both apply the same thresholds and the same reimbursement clock to all 5,570 municipalities and produce decisions at the level of the institution, not the town—which makes a municipality's own mortality shocks a less natural explanation for closure timing than the national cycle and the reimbursement squeeze (Section 2). None of this makes timing random, and I do not claim it is. I probe the condition in four ways below: closure timing is not strongly predicted by the local pre-closure conditions I can measure, though some pre-trend diagnostics are nonzero, so I do not treat these tests as proof of unconfoundedness; parallel trends survive event-study pre-periods and raw treated-versus-control plots; a synthetic difference-in-differences design that does not rest on never-treated parallel trends returns the same bounded null; and network-contamination checks confirm the controls are clean.

The PNASH sample, defined precisely. Two window definitions must be kept distinct. The primary causal sample, `PNASH_cycle_window`, is the 48 psychiatric closures (`tp_unid` = 07) that fall inside a PNASH inspection-cycle window—within ±1 year of either year of a two-year cycle. A stricter cut, `PNASH_exact_pm1`, keeps only the 37 closures within ±1 calendar year of the nearest single cycle year; the 11-closure gap is the 2014 cohort, two years from the 2011–2012 cycle. The main sample follows the cycle-window definition; a stricter exact ±1-year window yields similar mortality bounds (suicide +0.42 per 100,000, 95% CI $[-0.68, +1.51]$; self-harm +0.89, $[-0.75, +2.53]$), so the headline does not turn on the eleven distance-two closures (online appendix). The broader psychiatric sets (n = 41 under the statistical filter, n = 60 relaxing it) return as robustness in Section 6. This use of PNASH is a different cut from the F5 filter, which picks economically meaningful closures across the whole hospital universe; the failed-F5 diagnostic shows why gradual exits cannot be pooled with abrupt provider loss. An appendix event-level table documents each of the 48 closures—CNES, municipality, closure year, cycle distance, both window flags, beds, psychiatric volume, exposed catchment, baseline mortality, and the registry de-accreditation status and documented motive where recorded—so the timing and baseline characteristics of every event can be audited directly.

Table 4: PNASH identification and pre-treatment balance summary

Panel	Diagnostic	Value
Timing	PNASH psychiatric closures (primary sample)	48
Timing	Within ±1 year of a PNASH cycle	48
Timing	of which within ±1 calendar year of the nearest cycle date	37
Timing	Closures by cycle	2011-2012: 26; 2015-2016: 22
Predictors	Joint predictor test, closure year on covariates (F / p)	1.18 / 0.33
Balance	Log population std. diff. / p	-0.220 / 0.039
Balance	Suicide mortality std. diff. / p	-0.127 / 0.215
Balance	Self-harm mortality std. diff. / p	-0.273 / 0.008
Balance	Psychiatric admission episodes std. diff. / p	+0.301 / 0.006

Notes: Balance uses only observations strictly before exposure for treated municipalities. Predictor-test rows are closure-level regressions of closure year on standardized pre-closure predictors. Full closure-level timing and covariate tables are in the online appendix.

Does timing track local conditions? If areas with deteriorating mental health closed their hospitals earlier, timing would not be orthogonal to the outcome. I test this at the closure level. Regressing the closure year on the baseline conditions I can measure—psychiatric admissions, CAPS facilities, and the pre-closure volume decline—yields no joint predictive power (F-test $p = 0.25$, $N = 48$), and an early-versus-late closure split is likewise unpredicted ($p = 0.18$). A noisier test that adds per-catchment pre-closure outcome *trends* is less clean: two of the trend slopes are individually significant and their joint test is marginal ($p = 0.07$, $N = 34$), so I do not claim timing is wholly unrelated to pre-closure trajectories. What matters is that the panel pre-periods the estimator actually uses are flat, and the synthetic design below reweights controls to match each cohort's pre-closure path precisely, neutralizing this concern (online appendix).

The core assumption: parallel trends. The causal claim rests on one assumption above all: absent closure, exposed and never-treated municipalities would have tracked the same average mortality path. I assess it on pre-period event-study coefficients observed back to 2010, the early cohorts having real pre-periods only because the population denominators are backfilled (Section 3). The mortality pre-period coefficients are small and imprecise, the tests do not reject, and the first-stage pre-period is flat; raw treated-versus-control trajectories show no visible differential pre-closure mortality trend (online appendix). A flat pre-trend is not proof of parallel counterfactuals, and the pre-test is underpowered at this sample size. I report a HonestDiD relative-magnitude sensitivity (Rambachan and Roth, 2023), but at the per-100,000 scale of small municipalities the mortality series is too noisy for it to be informative: the robust set widens rapidly once any pre-trend extrapolation is allowed, so I do not lean on it for the mortality null (Section 5.4). The same machinery is decisive against ICSAP, where it confirms no bounded extrapolation rescues the effect; ICSAP fails the pre-trend check outright, which is why I decline to read it causally (Section 5.6).

No anticipation. Closures are quasi-public events with weeks to months of notice, not years, so I set the anticipation horizon to zero and let the pre-period coefficients stand as the implicit test. Re-normalizing the event study to $e = -3$, a base further from any pre-closure adjustment, leaves the suicide ATT at $+0.36$, essentially the headline value.

Pre-treatment balance. Table 4 summarizes balance for the PNASH psychiatric causal sample, using only pre-exposure observations for treated municipalities; the full table is in the online appendix. The balance tells the design what it can and cannot lean on. Flow-exposed municipalities have more psychiatric admission episodes before closure, exactly as the exposure rule implies, but pre-treatment suicide and travel burden are not statistically different at conventional levels. Self-harm and preventable hospitalizations differ modestly at baseline. The causal claim therefore rests on within-municipality event-study comparisons and pre-trend checks, not on cross-sectional equality of the two groups.

A second design, spillovers, and the limits. The event study identifies off never-treated parallel trends. Because that assumption is not testable, I add a design that does not require it. Synthetic difference-in-differences (Arkhangelsky et al., 2021) reweights control municipalities to match each closure cohort's pre-closure trajectory and is valid if *either* parallel trends *or* the synthetic weights hold. Across 4 cohorts and a donor pool of 5,460 never-treated municipalities, with a credible pre-treatment fit (RMSE/baseline = 0.24), it returns a suicide ATT of -0.22 per 100,000 (95% CI $[-1.80, +1.36]$), inside the event-study interval—so two designs with different identifying assumptions converge on the same bounded null (online appendix). SUTVA is the remaining concern: a closure can push patients onto a hospital that also serves an untreated control, contaminating it. Network-based checks that drop controls sharing a health-region proxy, a referral hub, the substitute hospitals that absorbed redirected flow, or a high flow-similarity link with treated catchments leave the null intact, in the paper's own network language (Section 6). Three honest limits remain. The design is not randomized; the sample is small (48 closures), so it bounds large effects rather than pinning a precise zero; and it observes inpatient care and mortality but not whether outpatient capacity absorbed displaced patients—community CAPS coverage is recorded only as facility presence, too coarse to test the substitution channel (Section 7.4). A cleaner instrument is also unavailable: the reimbursement freeze is national and the closing facilities are homogeneously SUS-funded psychiatric hospitals, leaving no cross-hospital exposure variation for a shift-share first stage (a reimbursement-vulnerability instrument yields a first-stage F near 4). Identification rests on the staggered design and its convergence with synthetic difference-in-differences, not on an excluded instrument.

4.3 Estimator

I estimate effects with the staggered event study of Sun and Abraham (2021), which holds up under the cohort-heterogeneous treatment effects that a closure sample spanning 2012–2023 invites. Let i index municipalities (flow-exposed and never-treated controls), t index years in $[2010, 2024]$, and g_i the year of the earliest closure to which i is exposed ($g_i = \infty$ for never-treated). With $\text{rel}_{i,t} = t - g_i$,

$$Y_{i,t} \;=\; \sum_{e \neq -1} \sum_{g} \delta_{g,e} \, \mathbf{1}\{g_i = g\} \, \mathbf{1}\{\text{rel}_{i,t} = e\} \;+\; \alpha_i \;+\; \gamma_t \;+\; \varepsilon_{i,t}, \tag{3}$$

with municipality fixed effects α_i, year fixed effects γ_t, and reference period $e = -1$. The error $\varepsilon_{i,t}$ collects municipality-year shocks to mortality—local labor-market conditions, other mental-health policy, year-to-year recording fluctuations—net of the municipality and year fixed effects; identification asks that closure timing be uncorrelated with these shocks, the economic case for which is the federal PNASH calendar (Section 2). The specification carries no time-varying covariates by design: admissions, travel burden, and the like are themselves outcomes of the closure, so conditioning on them would be a bad control. Following Sun and Abraham (2021), I aggregate the cohort-specific $\widehat{\delta}_{g,e}$ into the dynamic trajectory plotted in the event studies and into the simple post-period average reported as the ATT in the tables. Standard errors are clustered at the municipality level.

Weighting. For the rate outcomes the primary estimates are population-weighted, for two reasons: the policy-relevant quantity is the effect on the average exposed *person*, and population weights steady the per-capita rates of small municipalities. The unweighted (municipality-weighted) estimate sits alongside it, and both land on the same null (Section 5.4).

Alternative estimators and auxiliary diagnostics. The mortality null also holds under Callaway and Sant'Anna (2021) and Borusyak et al. (2024) in Section 6. Appendix analyses report auxiliary machine-learning, graph, documentary, and in-hospital outcome diagnostics. None of them enters treatment assignment, identification, or the headline mortality estimates.

5 Results

The results come in four steps. The full F5 sample shows that flow and distance pick out different exposed catchments. Travel-burden first stages then show that the flow-exposed municipalities had real utilization links to the closing providers, separately for the F5 sample and the PNASH psychiatric subset. The mortality estimates follow, on the PNASH closures alone. Nonpsychiatric

F5 closures and failed-F5 predecline closures close the section as contrast and diagnostic cases. The full robustness battery is in Section 6.

5.1 Flow versus distance in the full F5 closure sample

Start with measurement, on the full F5 sample of 60 economically meaningful closures. Table 3 shows the two rules pick out sharply different catchments. Of the 337 municipality–closure pairs either rule flags, 97 are flow-only and 196 are distance-only; just 44 are flagged by both. The disagreement, 293 of 337 pairs, is the paper's central evidence that geographic proximity is an incomplete proxy for exposure to provider loss in a referral network. The same table carries the specialty-aware distance benchmarks and flags the comparisons whose support is too thin to trust.

5.2 Pre-closure flows identify the exposed catchments

Before asking what happened to the exposed populations, I check that the flow rule is selecting municipalities with real admission links to the closing hospital, not ones that merely sit near it. Travel burden is the diagnostic. A post-closure change in travel distance can come from redirection to nearer providers, from the disappearance of long-distance admissions, or from a shift in admission composition; its sign and size speak to realized utilization, not to welfare.

In the full F5 sample, flow exposure gives a −5.84 km post-closure travel-burden ATT, against −3.63 km under the generic distance rule. In the PNASH psychiatric causal sample the flow first stage is larger, −7.09 km, while the distance-exposed series carries a strong pre-trend. Both patterns support the measurement claim and the choice to run the causal mortality design on the PNASH psychiatric sample. First-stage estimates for every sample are in the online appendix.

The sign of the first stage matters for how to read the mortality null. Travel *falls* after closure, because the long-distance admissions to the closing psychiatric hospital disappear or shift to nearer providers. The mechanical channel through which closures are usually feared to raise mortality—patients facing longer trips to care—does not operate here. Any acute mortality effect would have to run through the loss of the specialized inpatient option itself, not through added distance, which makes the absence of a mortality response easier to credit rather than harder.

5.3 Where inpatient psychiatric use goes after closure

Before turning to mortality, it helps to know how large the shock to inpatient psychiatric care actually is, and where the admissions go. Inpatient psychiatric admissions—admissions with a mental or behavioural primary diagnosis (ICD-10 F00–F99), counted at the municipality of residence across *all* hospitals—fall by −7.32 per 1,000 after closure (95% CI $[-11.38, -3.27]$), against a treated-group baseline of 10.7—a reduction of about 69%, with no pre-trend ($p = 0.34$; event-study figure in the online appendix).

The decomposition explains where the admissions go. Across the 30 closures with adequate flow support, exposed catchments averaged 596 psychiatric admissions per year to the closing hospital before closure; admissions to other hospitals do not rise to replace them, so total inpatient psychiatric admissions fall (Figure 3 and Table 5). The non-absorption is causal, not a trend artifact. Re-estimating the staggered event study with the outcome redefined as admissions at hospitals *other* than the closing one leaves them statistically unchanged (ATT 0.08 per 1,000, 95% CI $[-0.74, 0.91]$, against a pre-closure base of 3.32 and flat pre-trends), while the same specification on total admissions reproduces the 69% decline (online appendix). Other hospitals do not measurably absorb what the closing hospital removed; the loss is a net contraction of inpatient psychiatric capacity, not a hospital-to-hospital reallocation.

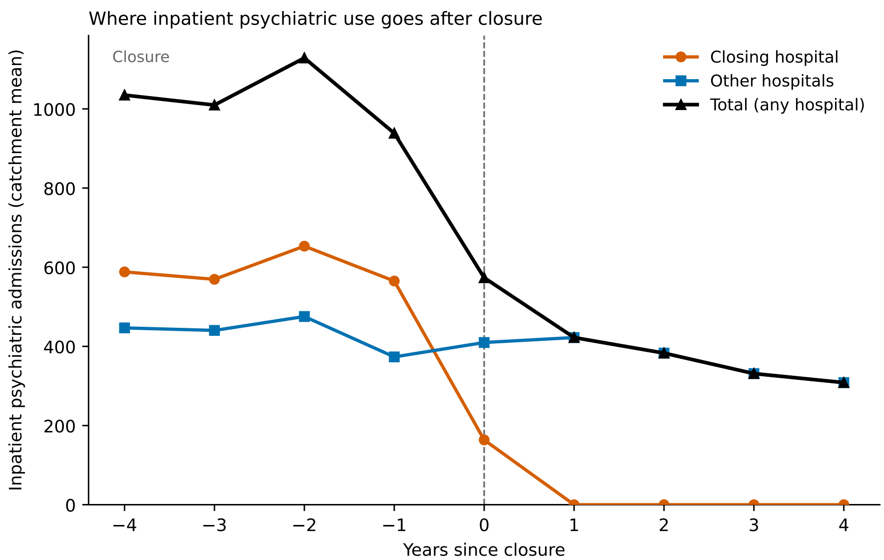

Figure 3: Where inpatient psychiatric use goes after closure. Catchment-level mean inpatient psychiatric admissions (ICD-10 F00–F99) by event time, across the 30 PNASH closures with flow support (municipalities sending $\geq 5\%$ of their pre-closure inpatient flow to the closing hospital). Admissions to the closing hospital collapse to zero at closure; admissions to other hospitals do not rise to compensate, so total inpatient psychiatric admissions fall and stay down. The design observes inpatient admissions only: "loss" means volume not observed in inpatient psychiatric care, not that residents receive no care.

The decline is a net loss of inpatient psychiatric *admissions*, not necessarily of all mental-health care. What disappeared was real capacity, not low-intensity custodial volume: admissions at the closing hospitals were no shorter in length of stay than at the hospitals that survived, and bed-days fell as steeply as admissions (66% against 69%, in the online appendix). Where that volume went is only partly identifiable from inpatient records. The plausible destination is the community and outpatient care the reform was building—the CAPS rollout that Dias and Fontes (2024) study, where hospitalizations also fall but outpatient care substitutes by design—yet the SIH does not observe CAPS production, except through the coarse baseline capacity controls in the appendix, and a residual of unmet need cannot be ruled out. What the data do establish is the negative: the

Table 5: Where inpatient psychiatric use goes after closure

Inpatient psychiatric admissions	Pre	Post	Change
To the closing hospital	596	0	-596
To other hospitals	430	379	-51
Total (any hospital)	1,026	379	-647

Notes: Catchment-level averages of inpatient psychiatric admissions (ICD-10 F00–F99), over event years -3 to -1 versus $+1$ to $+3$, for the 30 PNASH psychiatric closures with sufficient flow support. Exposed catchments are municipalities sending $\geq 5\%$ of their pre-closure inpatient flow to the closing hospital. The 596 admissions lost at the closing hospital are not picked up by other hospitals: admissions to other hospitals do not rise (they fall by 51), so total inpatient psychiatric admissions fall by about 63%. The lost volume is not observed in inpatient psychiatric care; the design observes inpatient admissions only, not outpatient (CAPS) use.

volume did not move to other hospitals. Its role here is to size the shock that follows: a contraction this large is what makes the mortality null informative rather than a quiet non-event.

5.4 Mortality after a large inpatient shock

Here I leave the measurement sample and restrict the causal analysis to the 48 PNASH-anchored psychiatric closures. The utilization result gives the mortality test its bite: the null that follows is not a quiet non-event after a weak first stage, but a bounded mortality response after a large loss of inpatient psychiatric care—the exact setting in which acute mortality should move, if it moves at all. The question is whether these psychiatric provider losses raised suicide or self-harm mortality in the municipalities that had relied on the closing hospitals. I measure two outcomes at the municipality of residence, suicide (ICD-10 X60–X84) and a broader self-harm measure that adds deaths of undetermined intent (Y10–Y34), each per 100,000 inhabitants. Population-weighted estimates are primary, because the policy-relevant quantity is the effect on the average exposed *person*, not the average exposed municipality; the municipality-weighted estimate is its unweighted complement.

Figure 4 plots the event-study trajectories and Table 6 reports the summary estimates. Population-weighted suicide mortality changes by +0.28 per 100,000 after closure (95% CI $[-0.72, +1.29]$), against a treated-group baseline of 5.01, with flat pre-period coefficients and a parallel-trends test that does not reject ($p = 0.26$). Self-harm is likewise indistinguishable from zero (+0.61, 95% CI $[-0.90, +2.12]$, $p = 0.53$). The municipality-weighted estimates are, if anything, negative, and stay statistically null. Across every specification the post-period trajectory sits near zero, with no upward drift large enough to show up in these data.

A non-rejecting pre-test can simply be underpowered, so I do not rest the null on it. Two things carry the weight instead. First, a synthetic difference-in-differences design that reweights controls to match each cohort's pre-closure path—and so does not require never-treated parallel trends—returns a suicide ATT of -0.22 per 100,000 (95% CI $[-1.80, +1.36]$), inside this interval (Section 4.2, online appendix). Second, the null survives every network-contamination exclusion of potentially spilled-over controls (Section 6). I also report a HonestDiD relative-magnitude sensitivity (Rambachan and Roth, 2023), but state plainly what it can and cannot do here: at the per-100,000 scale of small municipalities the mortality series is too noisy for relative-magnitude restrictions to be informative—the robust set widens quickly once any pre-trend extrapolation is allowed—so the bounded-null reading rests on the classical interval and the synthetic design, not on trend extrapolation. The same sensitivity is informative against ICSAP, where it confirms no bounded extrapolation can rescue the effect.

The null is bounded, not a precise zero. The upper end of the suicide interval, +1.29 per 100,000, is about a quarter of the treated-group baseline. The estimate rules out large acute increases—a one-quarter rise in exposed suicide would fall outside the interval—but it does not pin down a precise zero: effects smaller than that bound are beyond what 48 closures can resolve. The policy reading follows the same line. The data do not show the large acute mortality spike that would be visible at this precision; they cannot speak to subtle harms or to non-mortality welfare margins. Suicide and self-harm form a small, pre-specified outcome family, so multiple testing works in the null's favor rather than against it: any family-wise or false-discovery adjustment only widens these intervals, and wider intervals around estimates this close to zero cannot manufacture a significant effect where the unadjusted tests find none. The null is not specification-dependent: across 15 specifications spanning samples, estimators, threats, and inference, the suicide estimate stays a tight null (Section 6, Figure 5).

Table 6: Mortality estimates and scaled upper bounds

Outcome	ATT (95% CI) per 100,000	Baseline rate	Upper bound (% of baseline)	Upper bound (deaths/year)
Suicide	+0.28 [−0.72, +1.29]	5.01	+26%	38
Self-harm	+0.61 [−0.90, +2.12]	7.58	+28%	62

Notes: Population-weighted Sun–Abraham (2021) event-study ATTs on the PNASH psychiatric closure sample (48 closures and never-treated control municipalities, 2010–2023), with municipality and year fixed effects. Standard errors are clustered at the municipality level; the 95% confidence interval is the ATT ±1.96 standard errors. "Baseline" is the population-weighted pre-closure rate in treated municipalities. The two upper-bound columns translate the upper endpoint of the confidence interval into substantive units: as a share of the baseline rate, and as deaths per year across the exposed catchments (population ≈ 2.9 million). The robustness battery behind these estimates appears in Figure 5 and the online appendix.

Population versus municipality weighting. The point estimates differ in sign across the two weightings: population-weighted slightly positive, municipality-weighted slightly negative. That reflects an effect that, if present, differs between small and large exposed municipalities, not a fragile result. Both weightings give imprecise estimates around zero with intervals that exclude large increases; they differ in magnitude, not in the qualitative conclusion. I lead with the population-weighted estimate because it answers the policy question—harm to people, not to places—and report the municipality-weighted one so the reader can see the null is not an artifact of down-weighting small areas.

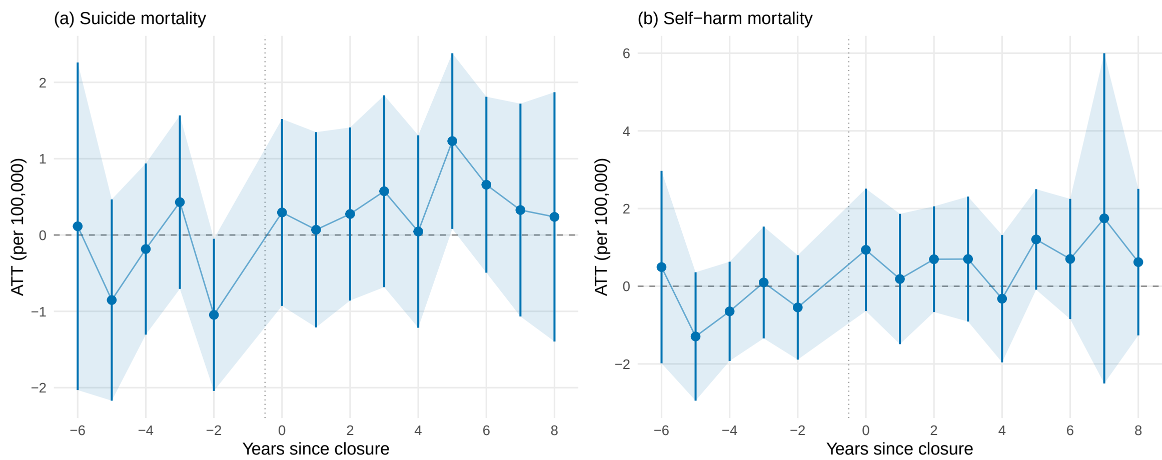

Figure 4: Mortality event studies. Population-weighted staggered Sun-Abraham event studies for (a) suicide and (b) self-harm mortality per 100,000, on the 48 PNASH-anchored psychiatric closures. Pre-periods are observed back to 2010. Both trajectories are flat before closure and remain indistinguishable from zero after. Bars are 95% cluster-robust confidence intervals.

5.5 Contrast and endogeneity diagnostics

General-hospital closures play no part in estimating the psychiatric mortality effect. They do a different job: testing whether the exposure-measurement problem is particular to psychiatric closures or shows up in hospital closures more broadly. The nonpsychiatric F5 sample holds 18 general-hospital closures and one hospital-day closure. Flow-defined and distance-defined catchments part ways here too (online appendix), which says that realized reliance and geographic proximity are distinct objects outside psychiatric reform as well. The travel-burden first stage is not read causally in this small, heterogeneous contrast sample.

General-hospital closures can also move measured travel in the opposite direction from psychiatric ones. A general closure may push patients toward farther substitutes, while a psychiatric closure can cut measured travel when long-distance specialized admissions vanish or shift to nearer providers. For that reason the nonpsychiatric rows are contrast evidence about exposure measurement, not a pooled mortality estimate.

The failed-F5 closures show why the F5 filter matters. These events clear the bed and volume screens but had already shed much of their patient volume before the formal closure date. Pooling them would blend abrupt provider loss with gradual institutional decline. The online appendix documents the pre-decline pattern and shows their travel-burden event studies failing pre-trend checks too, so I hold them out of the main design and use them only as a diagnostic.

5.6 A result I decline to read causally

A credible null has to be as honest about the outcomes the design cannot identify as about the one it can. Preventable hospitalizations (ICSAP), the canonical Brazilian access-to-care marker (Macinko and Harris, 2015; Rocha and Soares, 2010), do rise after closure in the exposed municipalities, by +1.99 per 1,000 in the post-period. Read naively, that looks like the access deterioration one would expect if closures pushed unmet need into preventable admissions. It is not identified.

The online appendix plots the trajectory. With pre-periods observed back to 2010, the pre-treatment coefficients are large and significantly positive—+2.03, +2.69, and +4.89 per 1,000 at three, four, and five years before closure—and the parallel-trends test rejects decisively ($p < 0.001$); a Rambachan-Roth sensitivity confirms that no bounded extrapolation of the pre-trend rescues the estimate. ICSAP was already climbing in the exposed municipalities well before any hospital closed, so the post-closure level is the continuation of a pre-existing trend, not a treatment effect. I had earlier mistaken this for a clean result when the pre-2015 data were missing and the pre-period looked empty; with the denominators backfilled to 2010, the trend is visible and the apparent effect dissolves.

I therefore report the ICSAP trajectory in full and treat it as non-identified. I do not bound it, do not read it causally, and do not let it carry weight in the paper's conclusions. The identified result is the mortality null; the ICSAP series is informative only as a description of where the exposed municipalities were already heading.

6 Robustness and diagnostics

Figure 5 puts the robustness battery in one picture. It plots the population-weighted suicide ATT across 15 specifications that vary the sample, the estimator, the controls, the inference, and the leading threats to the design. The point estimates fall in a narrow band—-0.22 to $+0.36$ per 100,000 across the design-based specifications—and every 95% confidence interval includes zero, with a single exception: the naive two-way fixed-effects benchmark, the one estimator known to be biased under staggered timing. The null is a feature of the data, not of a chosen specification. The rest of this section walks through the individual checks behind the curve; the full battery is in the online appendix.

Alternative psychiatric samples. The headline does not hang on the F5 pre-decline screen. The stricter sample of dedicated psychiatric hospitals (105 exposed municipalities) gives a population-weighted suicide ATT of +0.36 per 100,000 (95% CI $[-0.65, +1.37]$); the broader psychiatric sample that relaxes F5 (112 exposed) gives +0.34 ($[-0.62, +1.31]$). Both sit beside the primary +0.28 and exclude increases above roughly a quarter of baseline.

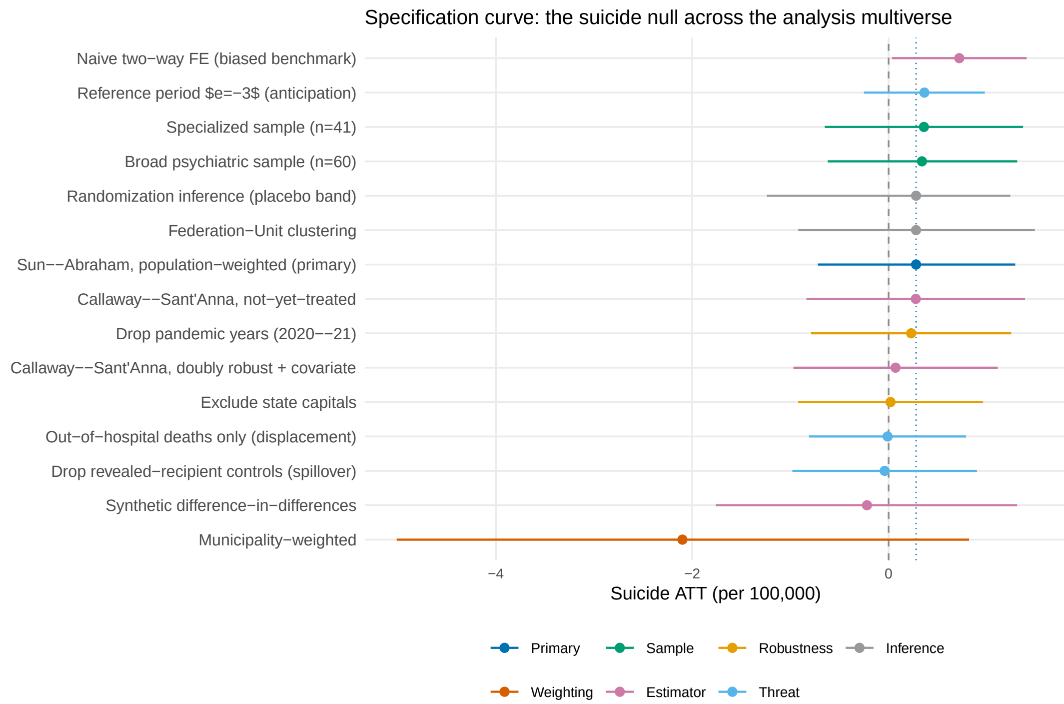

Figure 5: Specification curve for the suicide effect. Each row is one analytic specification; the point is its population-weighted ATT and the bar its 95% confidence interval, colored by the choice being varied (sample, weighting, estimator, robustness screen, identification threat, or inference). The dashed line marks zero and the dotted line the primary Sun-Abraham estimate. Every interval covers zero except the naive two-way fixed-effects benchmark (top), which is biased upward under staggered adoption and is shown only for contrast.

Alternative estimators. The null reproduces under Callaway and Sant'Anna (2021) and Borusyak et al. (2024). The one estimate that drifts is the naive two-way fixed-effects specification, +0.72 per 100,000—the upward pull expected when staggered treatment-effect heterogeneity contaminates the TWFE comparison, and the reason the design leans on Sun and Abraham (2021) to begin with. A Goodman-Bacon decomposition confirms the diagnosis (Goodman-Bacon, 2021): 99.6% of the TWFE weight falls on clean never-treated comparisons and only 0.3% on the forbidden already-treated ones, so the drift is heterogeneity weighting, not contaminated comparison groups. The null also holds under doubly-robust Callaway–Sant'Anna estimation with a not-yet-treated control group (+0.28 per 100,000) and under covariate adjustment (+0.07).

Pandemic years and capitals. Dropping 2020–21 leaves the suicide ATT at +0.23 per 100,000; the closures themselves sit outside the pandemic window by construction, and removing the pandemic from the control years does not move the estimate. Excluding the 27 state capitals—the main sites of within-municipality hospital dispersion, where the centroid distance approximation is weakest—gives +0.02 and shifts the first-stage travel ATT by less than 1%.

Exposure-rule sensitivity. The flow cutoff is not driving the null. Sweeping θ across $[0.01, 0.20]$ keeps the negative travel-burden first stage intact and never produces a positive mortality response. Psychiatric-specific denominators flag many more municipalities, since psychiatric admissions are sparse in the all-SUS denominator; these broader rules are stress tests, not a substitute for the main all-SUS reliance rule. The full set of exposure-variant estimates and an alternative z-score threshold variant are in the online appendix.

Network contamination and influential closures. A closure can push patients onto a hospital that also serves a control municipality, contaminating that control. Dropping controls that share a health-region proxy, a referral hub, a substitute hospital, or a high flow-similarity link with treated catchments leaves the suicide ATT close to the primary estimate. Removing one closure catchment at a time yields suicide ATTs between +0.02 and +0.59 per 100,000, so no single event carries the result. A sharper spillover test drops the 1,424 never-treated municipalities whose own admissions to the substitute hospitals rose after closure—the revealed recipients of redirected flow—and re-estimates on the 4,037 clean controls. Because spillover attenuates a true effect toward zero, dropping the spillover-exposed controls should raise the estimate if an effect were hidden; instead the suicide ATT is -0.04 (95% CI $[-0.98, +0.90]$), if anything slightly below the headline, and the two event-study trajectories nearly coincide (online appendix). The self-harm point estimate does rise under this control set but stays insignificant.

Mechanical displacement and outcome recording. Two threats specific to a mortality null get direct tests. When a psychiatric hospital closes, in-facility deaths mechanically leave the catchment's records, which could offset a true positive effect. Splitting deaths by place of occurrence and re-estimating on out-of-hospital deaths alone—the ones a closing hospital cannot mechanically absorb—leaves the suicide ATT at -0.01 per 100,000 (95% CI $[-0.81, +0.79]$), with the in-hospital component only 17% of pre-closure suicide deaths (online appendix). Differential death-recording is the other threat: the ill-defined-cause rate (ICD-10 R96–R99) does not jump at closure (+0.33 per 100,000 on a base of 57), so cause-coding error is a level the fixed effects absorb rather than something closure manufactures.

Inference. The null is not an artifact of understated standard errors. Clustering at the Federation Unit (27 clusters) widens the suicide interval to $[-0.92, +1.49]$; randomization inference over 500 placebo reassignments returns a two-sided p of 0.62 and a placebo interval of $[-1.24, +1.24]$. Both widen the uncertainty and both stay centered near zero. The data rule out large acute mortality increases of the magnitude in Table 6; they do not rule out smaller harms or non-mortality welfare losses.

7 Discussion

7.1 The measurement lesson from the full F5 closure sample

The portable contribution is the measurement result. The closure literature reads exposure off geographic proximity (Buchmueller et al., 2006; Avdic, 2016; Carroll, 2019; Joynt et al., 2015; Lindrooth et al., 2018). In the full F5 sample that rule misclassifies 293 of the 337 municipality–closure pairs, missing the patients who crossed borders for specialized care and flagging neighbors who never set foot in the facility. Run on the same closures, a distance-based design would have estimated effects on a substantially different, partly disjoint population, diluting the flow-dependent municipalities with the merely nearby. The patient-flow rule, checked against the travel-burden first stage and the utilization decomposition, instead picks out the municipalities whose realized use was tied to the closing hospital.

The correction is not particular to Brazil or to psychiatry. It needs three things: specialized care concentrated in regional hubs, so that proximity and dependence pull apart; origin–destination claims from a single or near-universal payer; and patients who cross administrative borders at scale. Those conditions hold across many single-payer and single-payer-like systems, and where they hold, a closure study that defines exposure by generic distance risks estimating effects on a population other than the one that used the hospital. How far off, and in which direction, is knowable in advance from the flow data, as the divergence map of Figure 1 shows.

The general-hospital contrast and the failed-F5 diagnostic mark the edges of the lesson. Nonpsychiatric F5 closures show the exposure problem is not a psychiatric-reform artifact. Failed-F5 closures show that not every formal closure date is an equally clean shock: when volume has already drained away before exit, the event blends provider loss with slow institutional decline. The F5 filter is part of the design, not a tidying step.

7.2 Mortality bounds among flow-exposed municipalities

The mortality estimates speak to municipalities with revealed pre-closure reliance on the closing psychiatric hospitals, and nowhere else. The null is not a non-event dressed up as a finding: in the same population inpatient psychiatric admissions fall by 69%, so the design captures a large, real disruption to inpatient psychiatric care—and still no sign of a large acute mortality increase follows it. Population-weighted suicide mortality changed by +0.28 per 100,000 (95% CI $[-0.72, +1.29]$), self-harm is similarly imprecise around zero, and parallel trends hold on the backfilled data.

The policy reading has to honor the bound. The estimates rule out increases above about a quarter of baseline suicide among exposed municipalities; they do not rule out smaller mortality effects, nor any of the welfare losses that do not end in a death certificate. The design sees acute municipality-level mortality in the SUS-using population. It does not see quality of life, caregiver burden, untreated morbidity, or the mortality of individual former patients, which would take record linkage the data do not support.

The two papers examine the same reform from opposite sides of the market. Dias and Fontes (2024) estimate the demand side: the CAPS rollout that added community outpatient capacity, which pulled psychiatric hospitalizations down, left mental-health mortality unchanged, and raised violent crime. This paper estimates the supply side: the removal of inpatient capacity through closure, among the municipalities revealed to depend on it. Read together, the papers separate two margins of the reform: expansion of community care and loss of inpatient providers; both reduce psychiatric inpatient use, and neither shows large mental-health mortality effects at the margins studied. Table 7 sets the treatments, exposure units, and estimands side by side.

Table 7: How this paper differs from Dias and Fontes (2024)

Paper	Treatment	Exposure unit	Main margin	Main estimand/contribution
Dias and Fontes (2024)	CAPS rollout	Municipality receiving CAPS	Community outpatient capacity expansion	Effects of community-care expansion on utilization, hospitalizations, mortality, and crime
This paper	Psychiatric hospital closure	Municipalities with revealed pre-closure reliance on the closing hospital	Provider loss / referral-network disruption	Flow-based exposure to hospital closures and acute mortality among exposed catchments

7.3 Why identification uses only within-event variation

A tempting alternative is to compare mortality across municipalities by their flow isolation, with no closure shock at all. That cross-section is uninformative here, and the reasons it fails are the reasons the within-event design is needed. Vital-registration coverage is worse in the rural North and Northeast, exactly the regions with the widest flow-versus-distance access gaps, so any cross-sectional access–mortality gradient is pulled toward zero by measurement quality that moves with the regressor. Cross-municipal comparisons also tangle flow isolation up with the whole regional health-system architecture. A closure shock with municipality fixed effects absorbs both; a cross-section absorbs neither. I therefore lean only on within-event variation, in keeping with a broader access-measurement literature (Doyle, 2011) where cross-sectional expenditure- or access-on-mortality regressions fail for the same reasons. The eleven cross-sectional specifications, which show that the auxiliary network diagnostics add no out-of-sample predictive content over distance, are in the replication notes.

7.4 Limitations

Five limitations bound the claims. The closure sample is modest (48 primary, 60 at its widest), so the null rules out large effects but not small ones, and fine-grained heterogeneity is underpowered; it does, though, survive Federation-Unit clustering and randomization inference (Section 6), so the modest sample is not manufacturing the null through understated standard errors. The analysis sees the SUS-using population only: private inpatient flows go unrecorded, and whether the flow-versus-distance gap looks the same in high-private-penetration regions is open. Mortality is the acute margin, which leaves the slower consequences of deinstitutionalization—chronic disease paths, caregiver burden, quality of life—outside the frame. The design observes inpatient care and mortality but not whether outpatient care absorbed the displaced volume: the only community-capacity measure available is the count of CAPS facilities per municipality-year, a coarse presence proxy that does not capture beds, residential capacity, or service production and whose national coverage ends in 2016. Splitting exposed catchments by whether they had any CAPS at baseline (21 versus 83 municipalities) shows no detectable differential in the suicide effect (online appendix), which is consistent with, but does not establish, outpatient substitution; I cannot test the mediation channel directly. Finally, the timing argument rests on the PNASH calendar and the reimbursement freeze, institution-level forces that are plausibly orthogonal to local demand conditional on the design checks but are not randomized; the natural-language motive classification that documents the closure narratives has internal but not yet external validation, so it does only robustness work. None of the five overturns the identified result. Each marks where it stops.

8 Conclusion

Who loses access when a hospital closes has two answers in a referral network, and they rarely coincide. Reading exposure off realized patient flows rather than off the map, I find the two rules disagree on 293 of the 337 municipality–closure pairs either one flags across 60 Brazilian closures: the places a distance rule misses had been crossing borders to the hospital that closed, and the places it wrongly flags never used it. Built from 179.4 million SUS admissions, the rule carries over to any provider-loss setting with administrative utilization data, and a first-stage drop in inpatient travel burden confirms it selects municipalities with real ties to the closing provider, not ones that merely sit nearby.

The two-step design asks first where inpatient use was concentrated before closure—flow-revealed exposure identifies the municipalities that actually depended on the closing beds—and then where that volume goes after. Applied to the 48 PNASH-anchored psychiatric closures, the displacement diagnostics show that other hospitals do not measurably absorb it: a staggered event study leaves their psychiatric admissions statistically unchanged (ATT 0.08 per 1,000, 95% CI $[-0.74, 0.91]$) even as total admissions fall 69%. Inpatient psychiatric admissions at exposed municipalities fall by about 69%—a net loss of inpatient psychiatric care, not a reshuffling across providers—yet suicide mortality changes by only +0.28 per 100,000 (95% CI $[-0.72, +1.29]$) and self-harm sits imprecisely around zero. The null is not fragile: it holds across 15 specifications and under a synthetic difference-in-differences design that does not lean on never-treated parallel trends. The one outcome that does rise, preventable hospitalizations, was already climbing well before any closure, so I report it without a causal reading. The estimate is a genuine bound—it rules out the visible mortality spike that critics of deinstitutionalization feared, on the suicide margin and at the precision 48 closures allow—not a precise zero.

The policy lesson is twofold and concrete. When a single-payer system plans or evaluates a closure, the population that will bear the loss is recoverable in advance from its own claims records, and a radius drawn on a map can target the wrong towns: over-serving nearby municipalities that never used the facility and overlooking distant ones that did. On the substantive question that has shadowed Brazil's psychiatric reform, the hospital closures did not, by this measure, produce the acute mortality catastrophe its opponents predicted for the municipalities that depended on the closing beds. That is a narrow reassurance, not a blanket one: it covers acute mortality, not the slower welfare margins, and the exposed catchments, not the discharged patients themselves.

Two extensions follow directly. The same flow-exposure rule transfers to other provider losses whose catchments cross borders—rural obstetric units, emergency departments, single-specialty clinics—and the gap between flow and distance can be mapped there before any design is run. And the question answered here at the municipality level becomes an individual one with record linkage: matching the discharged patients of a closing psychiatric hospital to vital records would replace the bound reported here with a direct estimate of former-patient mortality, the margin the aggregate data cannot reach. Read alongside the evidence on CAPS rollout, the two margins of the reform—the expansion of community care and the loss of inpatient providers—can finally be weighed side by side rather than conflated.

Declarations

Declaration of competing interest. The author declares no competing financial or personal interests that could have influenced the work reported in this paper.

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability. All data are public secondary administrative records. Inpatient admissions (SIH-RD), mortality (SIM), and establishment (CNES) microdata are distributed by DATASUS (https://datasus.saude.gov.br); municipal population, GDP, and boundary files are from IBGE (https://www.ibge.gov.br). No individual can be identified from these aggregated records. The code that downloads the raw files and reproduces every table and figure, with a numbered pipeline and environment specification, is available at https://github.com/darciogm/bitter-pills/tree/main/paper18-hospital-deserts.

CRediT author statement. **Darcio Genicolo-Martins:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing.

Generative AI disclosure. During the preparation of this work the author used AI-assisted tools for code scaffolding and language editing. The author reviewed and edited all content and takes full responsibility for the contents of the publication.

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